

Long-Term Outcomes of Medical Management vs Bariatric Surgery in Type 2 Diabetes

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IMPORTANCE Randomized clinical trials of bariatric surgery have been limited in size, type of surgical procedure, and follow-up duration.

OBJECTIVE To determine long-term glycemic control and safety of bariatric surgery compared with medical/lifestyle management of type 2 diabetes.

DESIGN, SETTING, AND PARTICIPANTS ARMMS-T2D (Alliance of Randomized Trials of Medicine vs Metabolic Surgery in Type 2 Diabetes) is a pooled analysis from 4 US single-center randomized trials conducted between May 2007 and August 2013, with observational follow-up through July 2022.

INTERVENTION Participants were originally randomized to undergo either medical/lifestyle management or 1 of the following 3 bariatric surgical procedures: Roux-en-Y gastric bypass, sleeve gastrectomy, or adjustable gastric banding.

MAIN OUTCOME AND MEASURES The primary outcome was change in hemoglobin A_{1c} (HbA_{1c}) from baseline to 7 years for all participants. Data are reported for up to 12 years.

RESULTS A total of 262 of 305 eligible participants (86%) enrolled in long-term follow-up for this pooled analysis. The mean (SD) age of participants was 49.9 (8.3) years, mean (SD) body mass index was 36.4 (3.5), 68.3% were women, 31% were Black, and 67.2% were White. During follow-up, 25% of participants randomized to undergo medical/lifestyle management underwent bariatric surgery. The median follow-up was 11 years. At 7 years, HbA_{1c} decreased by 0.2% (95% CI, −0.5% to 0.2%), from a baseline of 8.2%, in the medical/lifestyle group and by 1.6% (95% CI, −1.8% to −1.3%), from a baseline of 8.7%, in the bariatric surgery group. The between-group difference was −1.4% (95% CI, −1.8% to −1.0%; $P < .001$) at 7 years and −1.1% (95% CI, −1.7% to −0.5%; $P = .002$) at 12 years. Fewer antidiabetes medications were used in the bariatric surgery group. Diabetes remission was greater after bariatric surgery (6.2% in the medical/lifestyle group vs 18.2% in the bariatric surgery group; $P = .02$) at 7 years and at 12 years (0.0% in the medical/lifestyle group vs 12.7% in the bariatric surgery group; $P < .001$). There were 4 deaths (2.2%), 2 in each group, and no differences in major cardiovascular adverse events. Anemia, fractures, and gastrointestinal adverse events were more common after bariatric surgery.

CONCLUSION AND RELEVANCE After 7 to 12 years of follow-up, individuals originally randomized to undergo bariatric surgery compared with medical/lifestyle intervention had superior glycemic control with less diabetes medication use and higher rates of diabetes remission.

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Type 2 diabetes is a progressive multifactorial disease affecting an estimated 10% of the world's population, or more than 500 million adults.¹ In the US, type 2 diabetes accounts for \$237 billion in direct medical costs and \$90 billion in lost productivity annually, while care for people with diabetes accounts for 1 in 4 US health care dollars.² Several small randomized clinical trials (RCTs) and larger observational studies indicate that bariatric surgery is superior to medical and lifestyle therapies for treatment of type 2 diabetes.³⁻⁵ To date, RCTs have been limited in number, sample size, single site, operation type, severity of obesity, and follow-up duration. Thus, despite the growing body of evidence, most clinicians and payers do not recommend bariatric surgery for type 2 diabetes unless an individual has a body mass index (BMI) of 35 or higher,⁶ and less than 1% of those with a BMI of 35 or higher consider or pursue surgical treatment.⁷ Moreover, despite the recent advent of medications that achieve a degree of weight loss approaching that of some bariatric surgical procedures, these agents are costly, do not have proven long-term efficacy, and require long-term use to maintain weight loss.

The Alliance of Randomized Trials of Medicine vs Metabolic Surgery in Type 2 Diabetes (ARMMS-T2D) consortium pooled long-term observational results from 4 US single-center randomized trials to determine the efficacy, durability, and safety of bariatric surgery compared with medical/lifestyle treatment for type 2 diabetes.⁸ To our knowledge, this represents the largest pooled analysis with the longest follow-up to date. We previously reported results from ARMMS-T2D at 3 years, demonstrating that bariatric surgery is more effective and durable than medical/lifestyle intervention for type 2 diabetes remission, even in those with a BMI of 25 to less than 35.⁹ Herein, we report long-term follow-up of participants in ARMMS-T2D at the primary end point of 7 years, and up to 12 years, after randomization.

Methods

Study design

Randomized trials were conducted from May 1, 2007, to August 30, 2013, in the US at 4 centers: Cleveland Clinic (STAMPEDE), Joslin Diabetes Center/Brigham and Women's Hospital (SLIMM-T2D), University of Pittsburgh (TRIABETES), and University of Washington/Kaiser Permanente Washington (CROSSROADS).¹⁰⁻²⁰ Written consent was obtained from participants at each center, ethics approval was obtained to pool study results, and the pooled study analysis was registered at clinicaltrials.gov (NCT02328599). The protocol is provided in [Supplement 1](#). Race and ethnicity were included in each study because they may contribute to differential outcomes from the interventions. Participants self-identified their race and ethnicity based on fixed categories. Original study eligibility included diagnosis of type 2 diabetes, BMI of 27 to 45, and age of 18 to 65 years. Randomization protocol and operation types differed by site. Although medical/lifestyle interventions differed by site, they were all intensive compared with usual care and were based on the Diabetes Prevention

Key Points

Question What is the long-term durability of glycemic control and safety of bariatric surgery compared with medical/lifestyle management of type 2 diabetes?

Findings Bariatric surgery led to superior glycemic control compared with medical/lifestyle intervention (between-group difference in hemoglobin A_{1c} of 1.4% at 7 years and 1.1% at 12 years), with less diabetes medication usage and higher rates of diabetes remission.

Meaning These results combined with existing evidence support the use of bariatric surgery for treatment of type 2 diabetes in people with obesity.

Program and Look AHEAD interventions. Patients who underwent an operation had standard postoperative care, which included guidance on eating and activity, at least monthly, in the first year. Observational follow-up was conducted through July 2022. Data from extended follow-up were pooled and harmonized across sites for this analysis. Additional details are provided in [Supplement 2](#).

Primary and Secondary Outcomes

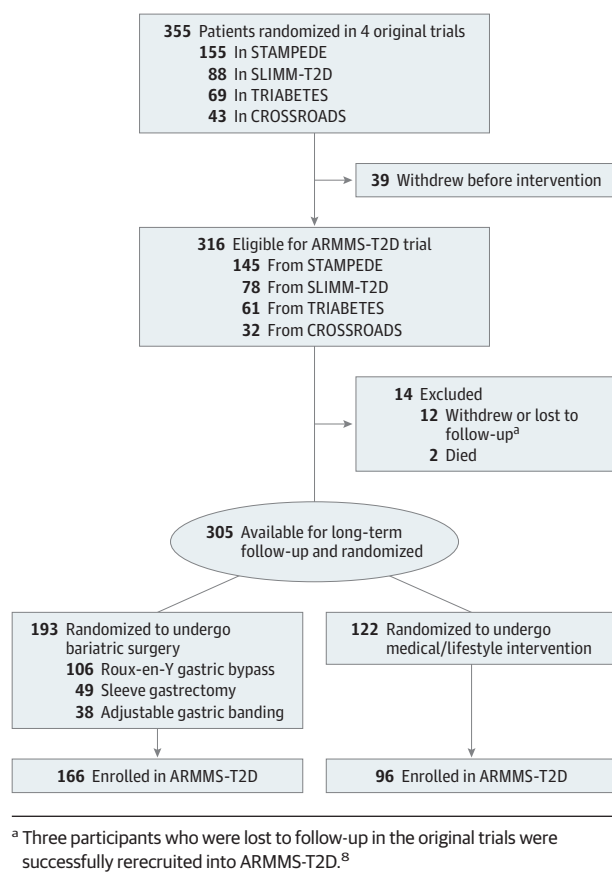
The primary outcome was the between-group difference in the percent change in HbA_{1c} from baseline to 7 years. Additionally, data up to 12 years are reported for participants who reached that follow-up point before study closure. Secondary outcomes included change over time in HbA_{1c}, HbA_{1c} less than 7.0%, and diabetes remission (HbA_{1c} <6.5% without diabetes medications for at least 3 months, assessed annually).^{21,22} Other secondary outcomes were between-group differences in change in weight, BMI, lipid levels, blood pressure, medication use, major adverse cardiovascular events, and microvascular complications. Main results are presented by randomized group as well as by surgical procedure type. For all outcomes, we hypothesized better results for bariatric surgery vs medical/lifestyle treatment. Throughout, to convert HbA_{1c} to mmol/mol, use the following equation: $10.93 \times \text{HbA}_{1c} - 23.50$.

Adverse events related to the original interventions were collected during annual study visits and at quarterly telephone calls by patient self-report, with verification by medical record review. Serious adverse events were defined as those resulting in death, a life-threatening event, hospitalization, significant disability, incapacity, or need for urgent medical or surgical intervention to prevent a serious outcome. Events with specific relevance, including revisional metabolic/bariatric procedures, severe hyperglycemia or hypoglycemia, and nutritional deficiencies are reported up to 12 years.

Statistical Analysis

The primary statistical analysis followed the intention-to-treat principle comparing the change in HbA_{1c} from baseline to 7 years between the 2 groups. Percent change in HbA_{1c} was assessed using a linear mixed-effect model, which included group, visit, group $\times$ visit interaction, site, and baseline HbA_{1c}.

Figure 1. Assembly of the Trials in the Alliance of Randomized Trials of Medicine vs Metabolic Surgery in Type 2 Diabetes (ARMMS-T2D)



as fixed effects, as well as a random intercept at the participant level. Least-square means were derived at each visit by group, and the group comparison was performed by applying appropriate linear contrast to the model. A per-protocol sensitivity analysis was also conducted to account for crossover from the medical/lifestyle group to the surgical group using the inverse probability weighting approach,²³⁻²⁵ where the model for the inverse probability weight assumed the current treatment depended on the previous treatments and outcomes. Missing clinical or laboratory data were recovered using the participants' electronic medical record, as described in Supplement 2. Death or loss to follow-up were assumed to be random censoring events, and the monotone missing data caused by these events were also considered missing at random. Further details about handling of missing data are provided in Supplement 2. Binary outcomes, such as diabetes remission and use of medications for diabetes, were summarized as percentages and analyzed using the generalized estimating equation approach controlling for site and baseline HbA_{1c}; standard errors of parameter estimates were obtained from the sandwich method. The results of secondary outcomes were not adjusted for multiple testing. Exploratory analyses were performed to investigate the subgroup of participants with BMI less than 35 using the same set of models as for the overall

cohort. All analyses were performed using RStudio (R Foundation) and SAS version 9.4 (SAS Institute).

Results

The 4 original trials included 355 individuals with type 2 diabetes who were randomized to undergo bariatric surgery vs medical/lifestyle intervention for management of type 2 diabetes (Figure 1).⁸ Randomization occurred at time of entry into the original trials; 39 individuals withdrew before the intervention (more than half due to dissatisfaction with assigned treatment group), leaving 316 eligible for the pooled analysis. Between the end of the original trials and the beginning of the pooled analysis, 9 participants withdrew their consent and 2 died. Of the 305 individuals available, 262 (86%) enrolled (Figure 1). Baseline characteristics of participants in the medical/lifestyle (n = 96) and surgical (n = 166) groups are shown in Table 1. Mean (SD) age was 49.9 (8.3) years, 68.3% of participants were women, 31% were Black, and 67.2% were White. Mean (SD) BMI was 36.4 (3.5) (eFigure 1 in Supplement 2) and 96 participants (36.6%) had a baseline BMI less than 35. Mean (SD) baseline HbA_{1c} was 8.5% (1.5%). Median (range) follow-up was 11 (7-15) years; some participants with later enrollment did not reach 12 years of follow-up. A comparison of patient characteristics with 7 and 12 years of follow-up is shown in eTable 1 in Supplement 2.

Absolute values for HbA_{1c} over 12 years are shown in Figure 2A. Despite higher baseline values, the bariatric surgery group had significantly lower HbA_{1c} levels than the medical/lifestyle group at all points after baseline ($P < .001$; Figure 2A). At 7 years, mean HbA_{1c} decreased to 8.0% from a baseline of 8.2% (difference, 0.2% [95% CI, -0.5% to 0.2%]) in the medical/lifestyle group and from 8.7% to 7.2% (difference, 1.6% [95% CI, -1.8% to -1.3%]) in the bariatric surgery group (Table 2). The between-group difference was -1.4% (95% CI, -1.8% to -1.0%; $P < .001$) at 7 years (Table 2) and -1.1% (95% CI, -1.7% to -0.5%; $P = .002$) at 12 years (eTable 2 in Supplement 2). At year 7, there were no significant differences in the improvements in mean HbA_{1c} from baseline between Roux-en-Y gastric bypass (difference, -1.7% [95% CI, -2.0% to -1.3%]) and sleeve gastrectomy (difference, -2.0% [95% CI, -2.6% to -1.5%]); HbA_{1c} improvement after adjustable gastric banding (-0.8% [95% CI, -1.3% to -0.2%]) was less than for sleeve gastrectomy ($P = .007$) and Roux-en-Y gastric bypass ($P = .03$) (Figure 2B). A per-protocol sensitivity analysis accounting for the 25% crossover from medical/lifestyle intervention to a surgical intervention showed a change in mean HbA_{1c} of 0.1% (95% CI, -0.5% to 0.7%) for the medical/lifestyle group and -1.4% (95% CI, -1.7% to -1.2%) for the bariatric surgery group at 7 years, with a between-group difference of -1.5% (95% CI, -2.1% to -0.9%; $P < .001$; eFigure 2A in Supplement 2).

Overall, 0.5% of participants in the medical/lifestyle group achieved remission of diabetes at 1 year, compared with 50.8% in the bariatric surgery group (Figure 3). Over time, the percentage achieving remission decreased in the bariatric surgery group, but remission rates remained higher

Table 1. Baseline Characteristics of Participants by Group (N = 262)

Characteristic	Medical/lifestyle (n = 96)	Bariatric surgery (n = 166)	Bariatric surgery type		
			Roux-en-Y gastric bypass (n = 89)	Sleeve gastrectomy (n = 41)	Adjustable gastric banding (n = 36)
Demographics, No. (%)					
Age, y	51.4 (6.8)	49.0 (9.0)	49.1 (9.0)	48.3 (7.7)	49.6 (10.3)
Sex					
Women	62 (64.6)	117 (70.5)	61 (68.5)	32 (78.0)	24 (66.7)
Men	34 (35.4)	49 (29.5)	28 (31.5)	9 (22.0)	12 (33.3)
Race					
Black	35 (36.5)	46 (27.7)	23 (25.8)	13 (31.7)	10 (27.8)
White	59 (61.5)	118 (71.1)	64 (71.9)	28 (68.3)	26 (72.2)
Other ^a	2 (2.1)	2 (1.2)	2 (2.3)	0	0
Anthropometrics, mean (SD)					
Waist, cm	113.7 (9.6) [n = 95]	115.0 (9.9)	116.1 (9.9)	113.3 (10.2)	114.5 (9.7)
Weight, kg	105.6 (15.5)	103.5 (15.3)	105.2 (15.3)	100.2 (16.7)	103.1 (13.0)
BMI	36.2 (3.4)	36.6 (3.6)	37.0 (3.4)	36.3 (4.2)	35.9 (3.2)
BMI <35	40 (41.7)	56 (33.7)	26 (29.2)	15 (36.6)	15 (41.7)
Systolic BP, mm Hg	129.7 (15.8)	134.4 (17.7)	135.0 (18.4)	135.8 (19.9)	131.6 (12.7)
Diastolic BP, mm Hg	79.5 (9.6)	80.4 (10.0)	80.7 (9.8)	81.9 (12.2)	78.2 (7.2)
Diabetes duration, y	8.8 (5.2)	8.3 (5.5)	8.8 (5.9)	7.8 (4.6)	7.5 (5.1)
Laboratory					
HbA _{1c} , mean (SD), %	8.2 (1.2)	8.7 (1.7)	8.7 (1.6)	9.4 (1.6)	8.2 (1.8)
HbA _{1c} <7.0%, No. (%)	11 (11.5)	20 (12.0)	9 (10.1)	0	11 (30.6)
Fasting glucose, mean (SD), mg/dL [n = 95]	156.5 (50.0)	172.0 (69.7)	171.0 (69.5)	172.1 (66.1)	174.6 (75.9)
Total cholesterol, mean (SD), mg/dL [n = 95]	172.6 (38.5)	179.6 (44.8)	176.1 (40.7)	191.2 (46.8)	174.9 (50.7)
HDL, mean (SD), mg/dL [n = 95]	44.3 (13.2)	42.9 (11.6)	43.6 (11.8)	45.5 (12.2)	38.1 (9.0)
LDL, mean (SD), mg/dL [n = 94]	96.3 (33.2)	100.3 (34.2) [n = 159]	96.3 (31.8) [n = 88]	110.8 (41.0)	97.8 (28.7) [n = 30]
Triglycerides, mean (SD), mg/dL [n = 95]	140.0 (92.5-221.5)	145.0 (103.0-231.8)	143.0 (100.0-239.0)	160.0 (120.0-214.0)	142.5 (94.0-251.5)
Serum creatinine, mean (SD), mg/dL [n = 95]	0.7 (0.2)	0.7 (0.2)	0.7 (0.2)	0.7 (0.2)	0.8 (0.2)
Urine albumin:creatinine ratio, mean (SD) [n = 74]	6.0 (4.0-14.5)	9.0 (4.0-22.5) [n = 119]	7.5 (3.0-28.0) [n = 62]	9.0 (7.0-22.0)	6.0 (3.8-12.2) [n = 16]
Medication use, No. (%)					
Statins	71 (74.0)	121 (72.9)	66 (74.2)	31 (75.6)	24 (66.7)
ACEi/ARB	65 (67.7)	108 (65.1)	63 (70.8)	25 (61.0)	20 (55.6)
ACEi	59 (61.5)	78 (47.0)	47 (52.8)	17 (41.5)	14 (38.9)
Insulin	36 (37.5)	82 (49.4)	46 (51.7)	17 (41.5)	19 (52.8)
β-Blockers	16 (16.7)	30 (18.2)	15 (16.9)	6 (14.6)	9/35 (25.7)
ARB	8 (8.3)	30 (18.1)	16 (18.0)	8 (19.5)	6 (16.7)

Abbreviations: ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; BP, blood pressure; HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol.

SI conversion factors: To convert hemoglobin A_{1c} (HbA_{1c}) to mmol/mol, multiply by 10.93 and subtract 23.50; glucose to mmol/L, multiply by 0.0555; cholesterol to mmol/L, multiply by 0.0259; triglycerides to mmol/L, multiply by 0.0113; creatinine to μmol/L, multiply by 88.4.

^a Including 2 patients identifying as Asian race and 2 patients who reported more than 1 race.

than in the medical/lifestyle group. At year 7, remission was 6.2% in the medical/lifestyle group compared with 18.2% in the bariatric surgery group (odds ratio, 3.4 [95% CI, 1.3-9.2]; $P = .02$; Figure 3), with the difference remaining statistically significant at 12 years ($P < .001$). Rates of remission were 24.5% in the Roux-en-Y gastric bypass subgroup, 15.2% in the sleeve gastrectomy group, and 8.9% in the adjustable gastric banding group; comparisons between surgical procedures were not statistically significant.

At 7 years, 26.7% of participants in the medical/lifestyle group had an HbA_{1c} less than 7.0%, compared with 54.1%

of participants in the bariatric surgery group (odds ratio, 3.2 [95% CI, 1.8-5.9]; $P < .001$; Table 2). Similar differences were observed using the threshold of HbA_{1c} less than or equal to 6.5% ($P = .002$; Table 2). Moreover, improved glycemic control in the bariatric surgery group was achieved using fewer medications. Rates of diabetes medication use at baseline were similar between the groups (Table 2; eFigure 3 in Supplement 2) and did not change significantly over time in the medical/lifestyle group ($P = .19$ at year 7 and 0.12 at year 12); in contrast, medication usage was reduced from 97.6% (162/166) at baseline to 38.0% (62/163) at year 1 in the

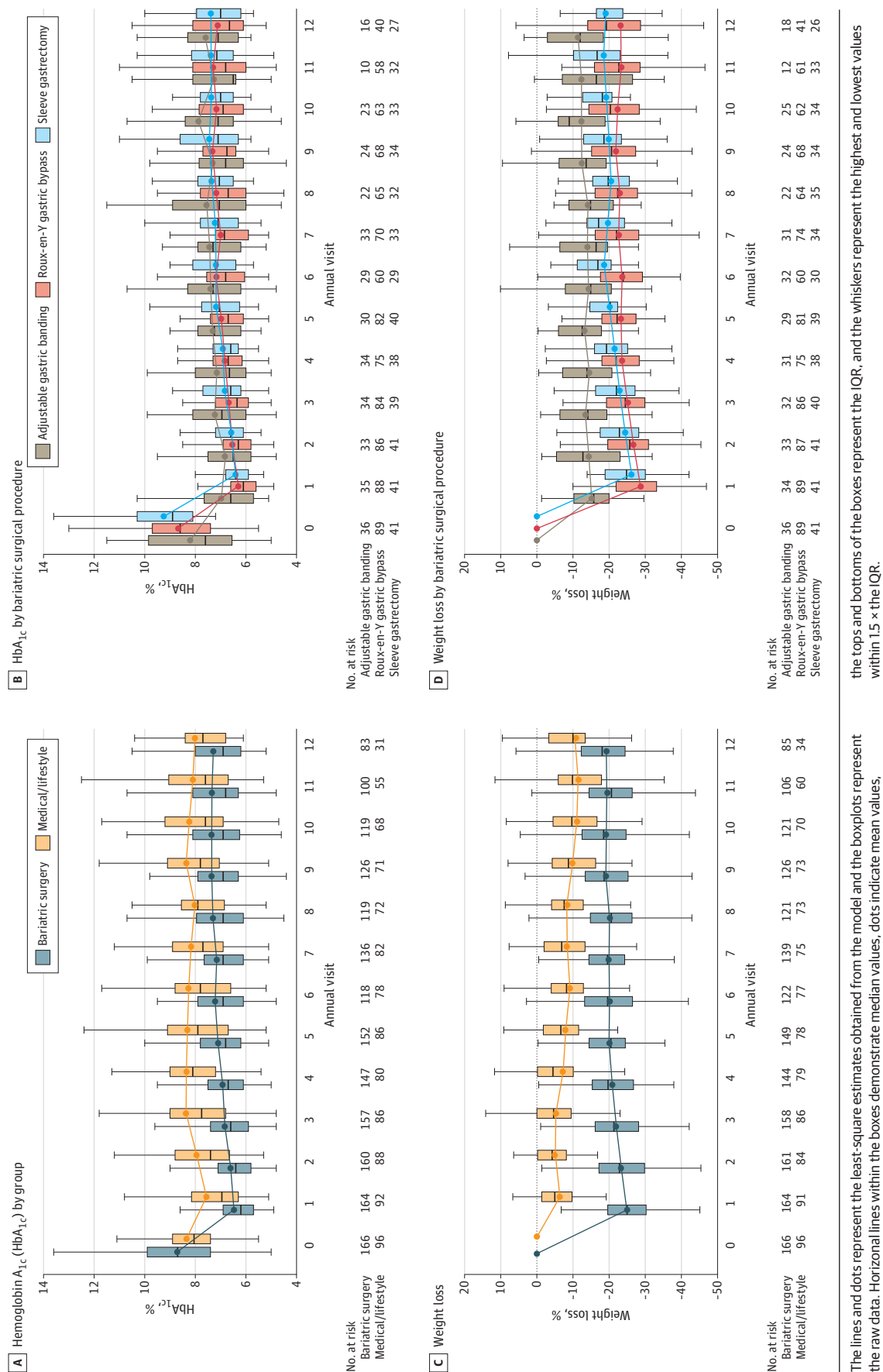
Figure 2. HbA_{1c} and Weight Loss by Group and Procedure Type

Table 2. Laboratory and Clinical Outcomes at Year 7 and Difference From Baseline^a

Outcome	Medical/lifestyle			Bariatric surgery			Group comparison	
	Baseline (n = 96)	Year 7 (n = 82)	Change (95% CI) ^b	Baseline (n = 166)	Year 7 (n = 136)	Change (95% CI)	Difference in change ^c	P value
Primary outcome								
HbA _{1c} , mean (SD), %	8.2 (1.2)	8.0 (1.8)	-0.2 (-0.5 to 0.2)	8.7 (1.7)	7.2 (1.4)	-1.6 (-1.8 to -1.3)	-1.4 (-1.8 to -1.0)	<.001
Secondary outcomes								
Fasting glucose, mean (SD), mg/dL ^d	156.5 (50.0)	144.6 (57.3)	-3.8 (-14.8% to 7.2%)	172.0 (69.7)	125.1 (47.0)	-14.1% (-22.0% to -6.3%)	-10.3% (-23.6% to 2.9%)	.13
Weight, mean (SD), kg	105.6 (15.5)	96.2 (16.6)	-8.3% (-10.5% to -6.1%)	103.5 (15.3)	83.6 (15.8)	-19.9% (-21.6% to -18.1%)	-11.6% (-14.3% to -8.9%)	<.001
SBP, mean (SD), mm Hg	129.7 (15.8)	128.7 (15.7)	-1.1% (-3.9% to 1.7%)	134.4 (17.7)	128.6 (15.3)	-3.4% (-5.6% to -1.2%)	-2.3% (-5.8% to 1.1%)	.19
DBP, mean (SD), mm Hg	79.5 (9.6)	74.6 (9.7)	-4.3% (-7% to -1.6%)	80.4 (10)	74.3 (10.4)	-6.0% (-8.1% to -3.8%)	-1.7% (-5.0% to 1.7%)	.32
LDL, mean (SD), mg/dL	96.3 (33.2)	97.6 (36.6)	5.5% (-3.3% to 14.3%)	100.3 (34.2)	103.1 (36.4)	10.8% (3.8% to 17.9%)	5.4% (-5.6% to 16.3%)	.34
HDL, mean (SD), mg/dL	44.3 (13.2)	52.0 (17.0)	20.5% (14.5% to 26.6%)	42.9 (11.6)	56.5 (16.5)	37.4% (32.6% to 42.3%)	16.9% (9.4% to 24.4%)	<.001
Total cholesterol, mean (SD), mg/dL	172.6 (38.5)	171.7 (41.6)	-0.7% (-5.6% to 4.1%)	179.6 (44.8)	181.4 (40.6)	4.9% (1.0% to 8.7%)	5.6% (-0.4% to 11.6%)	.07
Triglycerides, median (IQR), mg/dL	140 (92.5-221.5)	125 (88-178.3)	2.3% (-8.6% to 13.2%)	144 (103 to 231)	107 (82 to 142)	-19.0% (-27.8% to -10.2%)	-21.3% (-34.9% to -7.8%)	.002
Serum creatinine, mean (SD), mg/dL	0.7 (0.2)	0.8 (0.4)	9.5% (1.8% to 17.1%)	0.7 (0.2)	0.8 (0.2)	10.5% (4.4% to 16.7%)	1.1% (-8.4% to 10.5%)	.83
Urine albumin:creatinine ratio, median (IQR)	6 (4-12)	8 (4-13.5)	1.3 (0.9 to 1.9)	9 (4.5 to 23)	6 (4 to 10)	0.9 (0.7 to 1.2)	-0.4 (-1.0 to 0.1)	.10
Remission of diabetes, %	0.6	6.2	10.4 (0.4 to 279.4)	0.6	18.2	36.2 (1.9 to 699.0)	3.39 (1.25 to 9.17)	.02
Use of medications for diabetes, %	99.0	96.0	0.10 (0.01 to 3.27)	97.6	60.5	0.03 (0.01 to 0.11)	0.09 (0.03 to 0.24)	<.001
Oral/GLP1 only	57.3	40.0	0.53 (0.29 to 0.97)	47.0	44.5	0.74 (0.46 to 1.20)	0.98 (0.53 to 1.82)	.95
Insulin and/or oral/GLP1	41.7	56.0	1.93 (1.07 to 3.46)	50.6	16.0	0.18 (0.11 to 0.31)	0.13 (0.06 to 0.29)	<.001
HbA _{1c} <7.0%, %	11.7	26.7	2.77 (1.38 to 5.54)	15.5	54.1	6.42 (3.63 to 11.4)	3.22 (1.76 to 5.88)	<.001
HbA _{1c} <6.5%, %	8.3	17.3	2.30 (1.19 to 4.47)	12.0	37.7	4.44 (2.46 to 8.01)	2.89 (1.48 to 5.64)	.002

Abbreviations: DBP, diastolic blood pressure; GLP1, glucagon-like peptide 1; HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol; SBP, systolic blood pressure.

SI conversion factors: To convert hemoglobin A_{1c} (HbA_{1c}) to mmol/mol, multiply by 10.93 and subtract 23.50; glucose to mmol/L, multiply by 0.0555; cholesterol to mmol/L, multiply by 0.0259; triglycerides to mmol/L, multiply by 0.0113; creatinine to μmol/L, multiply by 88.4.

^a Mean (SD) or median (IQR) presented for baseline and year-7 data and least-square estimate (95% CI) presented for the changes and group comparisons.

^b Net change presented for HbA_{1c}, fold change (ie, ratio of year 7 data over baseline data) for urine albumin:creatinine ratio, relative changes (ie, percentage change from baseline) for other numeric outcomes, and odds ratios (7-year over baseline) for binary outcomes (remission, glycemic control, and use of medications for diabetes).

^c For numeric outcomes, the difference is defined as the 7-year change in the surgical group minus the 7-year change of the medical/lifestyle group; for binary outcomes, the difference is the odds ratio at year 7 (the odds of 7-year outcome in the surgery group over the medical/lifestyle group).

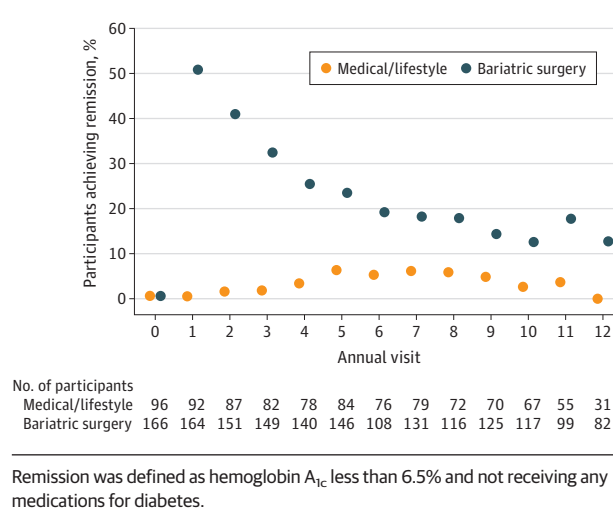
^d Only included fasting glucose measurements obtained at in-person study visits.

bariatric surgery group and remained significantly lower during follow-up compared with baseline (60.5% [72/119] at year 7; $P < .001$; Table 2; eFigure 3 in Supplement 2). Insulin usage after bariatric surgery was significantly lower than in the medical/lifestyle group (16% vs 56% at 7 years; $P < .001$; Table 2; eFigure 3 in Supplement 2) and the use of incretin/glucagon-like peptide 1 agonist medications was higher in the medical/lifestyle group across all annual visits ($P < .001$; eFigure 4 in Supplement 2).

Weight loss trajectories up to 12 years are shown in Figure 2C and D. Weight loss was significantly greater 7 years after bariatric surgery, with 8.3% weight loss (95% CI, 6.1%-10.5%) in the medical/lifestyle group and 19.9% weight loss (95% CI, 18.1%-21.6%) in the surgical group ($P < .001$; Figure 2C). Weight loss at 7 years was significantly greater for

Roux-en-Y gastric bypass than adjustable gastric banding (22.7% vs 14.0%; $P < .001$; Figure 2D), but did not differ significantly in sleeve gastrectomy (19.7% vs other surgical procedures. At 12 years, the bariatric surgery group continued to have superior weight loss (10.8% [95% CI, 8.2%-13.5%]) in the medical/lifestyle group vs 19.3% [95% CI, 17.3%-21.3%]) in the bariatric surgery group; $P < .001$; Figure 2C). At 7 years, a BMI less than or equal to 25, indicating nonobesity, was achieved in 2.7% of participants in the medical/lifestyle group and 14.4% in the bariatric surgery group. At 12 years, these rates were 0% in the medical/lifestyle group and 15.3% in the bariatric surgery group. eFigure 2B in Supplement 2 shows a comparison of the intention-to-treat and per-protocol analyses of weight loss by group over time. At year 7, weight loss was 5.6% for the medical/lifestyle group and 20.4% for the

Figure 3. Diabetes Remission



surgical group from the per-protocol analyses. At year 12, the per-protocol weight loss was 7.7% for the medical/lifestyle group and 19.4% in the bariatric surgery group after accounting for crossovers.

There were no significant differences in systolic blood pressure or low-density lipoprotein (LDL) cholesterol between the groups at 7 years (Table 2; eFigure 5A and 5C in Supplement 2). However, high-density lipoprotein (HDL) cholesterol was significantly higher ($P < .001$; eFigure 5B in Supplement 2) and triglycerides were significantly lower ($P < .001$; eFigure 5D in Supplement 2) over time in the bariatric surgery group. At year 7, the increase in HDL from baseline was greater for bariatric surgery (20.5% vs 37.4%; difference, 16.9% [95% CI, 9.4%-24.4%]; $P < .001$; Table 2). The percent change in triglycerides was 2.3% for medical/lifestyle and -19.0% for bariatric surgery at 7 years ($P = .002$). There were no significant changes in either serum creatinine or urine albumin-to-creatinine ratio between the groups (Table 2).

Results of an exploratory analysis of the 96 participants with a BMI of 27 to less than 35 vs those with a BMI of 35 or greater at randomization is shown in eFigure 6 in Supplement 2. The bariatric surgery group had significantly lower HbA_{1c} levels than the medical/lifestyle group at all points ($P < .001$; eFigure 6A in Supplement 2). At 7 years, the reduction in HbA_{1c} in the lower BMI surgical subgroup was significantly greater than in the lower BMI medical/lifestyle group (-0.4% vs -1.5%; difference, -1.2% [95% CI, -1.8% to -0.5%]; eFigure 6A in Supplement 2). The reduction in HbA_{1c} in the higher BMI group was -0.1% for the medical/lifestyle group and -1.6% for the surgical group (difference, -1.5% [95% CI, -2.1% to -1.0%]), which was not different from the lower BMI group ($P = .40$). Likewise, percent weight loss at year 7 in the subgroup of participants with a BMI less than 35 was significantly greater in the bariatric surgery group than the medical/lifestyle intervention group (5.6% vs 20.4%; difference, 14.8% [95% CI, 10.8%-18.8%]; $P < .001$; eFigure 6B in Supplement 2). Percent weight loss at year 7 in the higher BMI group

was 10.1% for the medical/lifestyle group and 19.3% for the bariatric surgery group, and the difference was 9.2% (95% CI, 5.6%-12.9%), which was significantly different from the lower BMI group ($P = .03$).

Deaths, major cardiovascular adverse events, and other major adverse events are shown in Table 3. There were 4 deaths over 12 years: 2 in the medical/lifestyle group (gunshot injury, disability from strokes leading to death) and 2 in the bariatric surgery group (cardiac event, COVID-19). Cardiovascular and other adverse events were similar between the groups except for fractures, anemia (hemoglobin <11.5 g/dL), and low iron (<59 µg/dL), which were more common in the bariatric surgery group (Table 3; eTable 3 in Supplement 2). The bariatric surgery group had significantly lower hemoglobin and higher vitamin B₁₂ and vitamin D levels than the medical/lifestyle group (eTable 4 in Supplement 2). Two of 96 participants (2.1%) in the medical/lifestyle group initiated kidney dialysis during follow-up compared with none in the bariatric surgery group; 5 of 96 participants (5.2%) in the medical/lifestyle group compared with 2 of 166 (1.2%) in the bariatric surgery group experienced retinopathy (Table 3).

There were more gastrointestinal events, such as abdominal pain, dumping syndrome, and nausea/vomiting, in the bariatric surgery group than the medication/lifestyle group (eTable 5 in Supplement 2). Procedures (surgical, endoscopic) during long-term follow-up are shown in eTable 6 in Supplement 2; there were no significant differences in procedures between the groups, except the combination of crossovers, conversions, and revisions, which were more common in the medical/lifestyle group. Specifically, 24 of 96 participants (25%) in the medical/lifestyle group underwent bariatric surgery (crossover) during follow-up (8 underwent Roux-en-Y gastric bypass, 15 underwent sleeve gastrectomy, and 1 underwent adjustable gastric banding) at a median (range) time of 4.5 (0.4-9.8) years (eTable 7 in Supplement 2). In the bariatric surgery group, 15 participants (9%) underwent conversion or revisional operations, with 7 adjustable gastric banding removals without further procedure, 4 adjustable gastric banding conversions to Roux-en-Y gastric bypass, 1 adjustable gastric banding conversion to sleeve gastrectomy, and 3 sleeve gastrectomy revisions to Roux-en-Y gastric bypass (2 for acid reflux, 1 for chronic fistula).

Discussion

This study reports results from the largest and longest follow-up to date of individuals with type 2 diabetes and overweight/obesity randomized to receive medical/lifestyle vs surgical treatments. In this pooled analysis of patients originally enrolled in 4 single center RCTs, at 7 years and up to 12 years of follow-up from randomization, bariatric surgery was more effective and resulted in long-term improvement for glycemic control while using fewer medications and weight loss. In parallel, increased rates of diabetes remission and reductions in medication use (including insulin) were observed in surgically treated participants. Bariatric surgery

was also more effective in improving HDL cholesterol and triglycerides at 7 years and up to 12 years of follow-up. These advantages should be balanced against the increased risk of nutritional deficiencies (anemia), gastrointestinal adverse events, and bone fractures in long-term follow-up after bariatric surgery.

Recent studies have highlighted that remission of diabetes is an important and achievable goal, particularly in early stages.²⁶ These data demonstrate that bariatric surgery is superior to medical/lifestyle interventions, with remission achieved in 51% of surgically-treated patients at 1 year and 18% at 7 years, compared with 0.5% of patients in the medical/lifestyle group at 1 year and 6% at 7 years.⁹ Participants included in the analyses had long-standing and suboptimally controlled type 2 diabetes (mean duration of diabetes, 8 years; mean HbA_{1c} at randomization, 8.5%). By contrast, the medical/lifestyle interventions, modeled after the Diabetes Prevention Program²⁷ and Look AHEAD trials,²⁸ were not sufficient to promote long-term improvements in glycemic control or achieve diabetes remission, despite modest weight loss. The relatively low remission rate at 7 years in the medical/lifestyle group is in contrast to the 2-year results from the DiRECT trial.²⁹ However, by design, participants in that study had a shorter duration of type 2 diabetes (mean, 3.1 years) and were not treated with insulin, and follow-up in that study was shorter. The reduction in remission over time after bariatric surgical procedures has also been observed in other studies, potentially related to weight regain, resolution of negative calorie balance, and progressive loss of β -cell function over time. Hence, even among patients who experience postoperative diabetes remission, continued surveillance for relapse is warranted. Even if relapse occurred, participants in the bariatric surgery group continued to experience better glycemic control while using fewer medications. Moreover, even relatively short-term remission has shown benefits for diabetes-related complications in other studies; risk of microvascular disease has been estimated to be reduced by 19% for each year of achieved remission.³⁰

Among RCTs of surgical vs medical/lifestyle management for type 2 diabetes, this analysis reports the longest period of active observation, with follow-up through 12 years after randomization. Beyond previous individual reports from the ARMMS-T2D consortium, the longest-term RCTs that reported comparison of surgical to medical/lifestyle management of type 2 diabetes were studies by Ikramuddin et al (5 years; 120 participants)⁴ and Mingrone et al (10 years; 60 participants).⁵ The study by Ikramuddin et al reported achievement of a composite outcome (HbA_{1c} <7.0%, LDL <100 mg/dL, and systolic blood pressure <130 mm Hg) in 23% of 57 patients who underwent a Roux-en-Y gastric bypass vs only 4% of 56 patients who underwent medical/lifestyle interventions. Likewise, the single-site RCT of Roux-en-Y gastric bypass vs biliopancreatic diversion vs conventional medical treatment from Mingrone et al reported diabetes remission at 10 years in 37.5% of 40 patients who underwent an operation compared with 5.5% of 18 patients in the medical group. Similar to ARMMS-T2D, these studies observed a waning over time of surgical benefits on diabetes metrics.

Table 3. Major Adverse Events and Events of Interest Through 12 Years

Event	No. (%)	
	Medical/lifestyle (n = 96)	Bariatric surgery (n = 166)
Death ^a	2 (2.1)	2 (1.2)
Cardiovascular		
Coronary revascularization	7 (7.3)	15 (9)
Myocardial infarction	4 (4.2)	10 (6)
Unstable angina	2 (2.1)	4 (2.4)
Significant arrhythmia	4 (4.2)	7 (4.2)
Heart failure	1 (1)	5 (3)
Stroke/transient ischemic attack	3 (3.1)	5 (3)
Peripheral arterial disease	0	2 (1.2)
Venous thromboembolism	2 (2.1)	1 (0.6)
Metabolic		
Severe hypoglycemia	7 (7.3)	11 (6.6)
Diabetic ketoacidosis	1 (1)	0
Gastrointestinal		
Gastric/anastomotic ulcer	2 (2.1)	10 (6)
Bowel obstruction	1 (1)	3 (1.8)
Gastrointestinal leaks	0	1 (0.6)
Gallstones/cholecystitis	3 (3.1)	9 (5.4)
Pancreatitis	1 (1)	3 (1.8)
Alcohol-associated cirrhosis ^b	0	2 (1.2)
Kidney		
Kidney stones	2 (2.1)	11 (6.6)
Initiation of dialysis	2 (2.1)	0
Ocular		
Retinopathy	5 (5.2)	2 (1.2)
Blindness	1 (1)	0
Blood transfusion		
For anemia	3 (3.1)	20 (12)
For gastrointestinal bleeding	2 (2.1)	5 (3)
Miscellaneous		
Fracture	5 (5.2)	22 (13.3)
Cancer ^c	4 (4.2)	9 (5.4)
Suicide attempt	0	1 (0.6)

^a The 2 deaths in the medical group were caused by gunshot injury and disability following strokes and the 2 deaths in the surgical group were caused by a cardiac event and COVID-19.

^b One patient had metabolic dysfunction-associated steatohepatitis (MASH) with bridging fibrosis at the time of Roux-en-Y gastric bypass. The participant's postoperative course was complicated with alcohol use disorder and the second liver biopsy 7 years after Roux-en-Y gastric bypass showed progression to cirrhosis. The second patient had simple hepatic steatosis (without MASH and fibrosis) on the initial liver biopsy at the time of Roux-en-Y gastric bypass and the postoperative course was complicated with decompensated alcohol-associated cirrhosis, which was confirmed with a second liver biopsy 8 years after Roux-en-Y gastric bypass.

^c Excluding nonmelanoma skin cancer.

Beyond glycemia, diabetes care goals extend to control of blood pressure and lipids and prevention of diabetes-related complications. Although no differences were observed between the groups in blood pressure or LDL cholesterol, bariatric surgery was superior at raising HDL and lowering triglycerides, which is consistent with previous smaller RCTs with

shorter follow-up.³¹ Although differences in blood pressure were not present, differential antihypertensive medication use between the groups may have confounded the results. A 2018 RCT comparing surgical vs medical/lifestyle interventions dedicated to blood pressure as the primary outcome reported surgical superiority in this domain.³²

Traditionally, the minimum BMI threshold for use of bariatric surgery among people with type 2 diabetes is 35.³³ In an exploratory analysis, reductions in HbA_{1c} were similar among participants randomized to the surgical group with baseline BMI at or above 35 or less than 35, and the difference in percent weight loss between the groups was greater in those with BMI less than 35. Data from other RCTs have also shown that bariatric surgery can be effective for type 2 diabetes treatment for those with BMI less than 35.³¹ These data support the newer standards for use of bariatric surgery among patients with a lower BMI and type 2 diabetes, such as those recommended by the Diabetes Surgery Summit and joint international surgical societies.^{31,34}

Anemia and fractures were more common after bariatric surgery. Micronutrient deficiencies may contribute to the higher fracture rate after bariatric surgery, and the potential for these and other deficiencies should be proactively monitored and measured life-long in patients after undergoing bariatric surgical procedures. Similar to other studies with shorter follow-up,³¹ gastrointestinal adverse events were higher after bariatric surgery.

Strengths of this analysis include inclusion of trials from 4 different sites, which may increase generalizability compared with the single-site composition of most previous RCTs.³¹ At baseline, more than one-third of participants had a BMI of less than 35, whereas most previous studies of bariatric surgery focused on people with a BMI of 35 or greater. Three bariatric surgical procedures were examined and data on sleeve gastrectomy, which is now the most commonly performed bariatric surgery operation worldwide, are included. More than one-fourth of the study participants were of racial and ethnic minority groups, which is relatively diverse compared with other studies of bariatric surgery.³¹ Additionally, the sample size and length of follow-up is substantially greater than in any previous single RCT.

Limitations

This study has several limitations. First, all included trials were open-label. Second, there were differences in original trial protocols and lack of uniform assessment for some diabetes complications, including retinopathy. Third, treatments examined in different trials were not identical. Randomization ratios were not all 1:1 (because some trials had 2 surgical groups). Trial populations may have had different distributions of effect modifiers or outcome predictors, so a pooled analysis such as this could create bias, or a form of confounding induced by the original trials. There may also have been selective dropout over time in the medical/lifestyle group; for example, those who were doing better from a health perspective may have participated longer. Fourth, there were different enrollment timelines, resulting in different numbers of participants with different lengths of follow-up.

Fifth, the assumption that data are missing at random was untestable with this study design. The per-protocol analysis relied on strong assumptions about the relationship between the (time-varying) treatment and the outcome in the model, the absence of uncontrolled time-varying confounding, and the specification of the model for the inverse probability weight. Sixth, this analysis was not powered to detect differences among the 3 surgical procedures for primary outcomes or to examine the risk of major adverse cardiovascular events, microvascular events, cancer, or mortality. Seventh, bariatric surgical procedures changed over the course of follow-up (eg, reduction in use of adjustable gastric banding). Eighth, increased use of incretin medications occurred in parallel in both groups during follow-up. As a result, changes in use of this class of medication over time could have influenced between-group differences in outcomes. Ninth, consistent with other bariatric surgery studies, most participants were women.

Conclusions

At 7 to 12 years of follow-up, individuals originally randomized to undergo bariatric surgery, compared with medical/lifestyle intervention, had superior glycemic control with less diabetes medication usage and higher rates of diabetes remission.

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REFERENCES

- Sun H, Saeedi P, Karuranga S, et al. IDF Diabetes Atlas: global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Res Clin Pract*. 2022;183:109119. doi:10.1016/j.diabres.2021.109119
- American Diabetes Association. Economic costs of diabetes in the US in 2017. *Diabetes Care*. 2018; 41(5):917-928. doi:10.2337/dci18-0007
- Schauer PR, Bhatt DL, Kirwan JP, et al; STAMPEDE Investigators. Bariatric surgery versus intensive medical therapy for diabetes: 5-year outcomes. *N Engl J Med*. 2017;376(7):641-651. doi:10.1056/NEJMoa1600869
- Ikramuddin S, Korner J, Lee WJ, et al. Lifestyle intervention and medical management with vs without Roux-en-Y gastric bypass and control of hemoglobin A1c, LDL cholesterol, and systolic blood pressure at 5 years in the Diabetes Surgery Study. *JAMA*. 2018;319(3):266-278. doi:10.1001/jama.2017.20813
- Mingrone G, Panunzi S, De Gaetano A, et al. Metabolic surgery versus conventional medical therapy in patients with type 2 diabetes: 10-year follow-up of an open-label, single-centre, randomised controlled trial. *Lancet*. 2021;397 (10271):293-304. doi:10.1016/S0140-6736(20)32649-0
- Gebran SG, Knighton B, Ngaage LM, et al. Insurance coverage criteria for bariatric surgery: a survey of policies. *Obes Surg*. 2020;30(2):707-713. doi:10.1007/s11695-019-04243-2
- Aminian A, Nissen SE. Success (but unfinished) story of metabolic surgery. *Diabetes Care*. 2020;43 (6):1175-1177. doi:10.2337/dci20-0006
- Simonson DC, Hu B, Arterburn DE, et al. Alliance of Randomized Trials of Medicine vs Metabolic Surgery in Type 2 Diabetes (ARMMs-T2D): study rationale, design, and methods. *Diabetes Obes Metab*. 2022;24(7):1206-1215. doi:10.1111/dom.14680
- Kirwan JP, Courcoulas AP, Cummings DE, et al. Diabetes remission in the Alliance of Randomized Trials of Medicine Versus Metabolic Surgery in Type 2 Diabetes (ARMMs-T2D). *Diabetes Care*. 2022;45(7):1574-1583. doi:10.2337/dc21-2441
- Schauer PR, Kashyap SR, Wolski K, et al. Bariatric surgery versus intensive medical therapy in obese patients with diabetes. *N Engl J Med*. 2012; 366(17):1567-1576. doi:10.1056/NEJMoa1200225
- Schauer PR, Bhatt DL, Kirwan JP, et al; STAMPEDE Investigators. Bariatric surgery versus intensive medical therapy for diabetes-3-year outcomes. *N Engl J Med*. 2014;370(21):2002-2013. doi:10.1056/NEJMoa1401329
- Courcoulas AP, Goodpaster BH, Eagleton JK, et al. Surgical vs medical treatments for type 2 diabetes mellitus: a randomized clinical trial. *JAMA Surg*. 2014;149(7):707-715. doi:10.1001/jamasurg.2014.467
- Courcoulas AP, Belle SH, Neiberg RH, et al. Three-year outcomes of bariatric surgery vs lifestyle intervention for type 2 diabetes mellitus treatment: a randomized clinical trial. *JAMA Surg*. 2015;150 (10):931-940. doi:10.1001/jamasurg.2015.1534
- Courcoulas AP, Gallagher JW, Neiberg RH, et al. Bariatric surgery vs lifestyle intervention for diabetes treatment: 5-year outcomes from a randomized trial. *J Clin Endocrinol Metab*. 2020;105 (3):866-876. doi:10.1210/clinem/dgaa006
- Halperin F, Ding SA, Simonson DC, et al. Roux-en-Y gastric bypass surgery or lifestyle with intensive medical management in patients with type 2 diabetes: feasibility and 1-year results of a randomized clinical trial. *JAMA Surg*. 2014;149(7): 716-726. doi:10.1001/jamasurg.2014.514
- Ding SA, Simonson DC, Wewalka M, et al. Adjustable gastric band surgery or medical management in patients with type 2 diabetes: a randomized clinical trial. *J Clin Endocrinol Metab*. 2015;100(7):2546-2556. doi:10.1210/jc.2015-1443
- Simonson DC, Halperin F, Foster K, Vernon A, Goldfine AB. Clinical and patient-centered outcomes in obese patients with type 2 diabetes 3 years after randomization to Roux-en-Y gastric bypass surgery versus intensive lifestyle management: the SLIMM-T2D study. *Diabetes Care*. 2018;41(4):670-679. doi:10.2337/dci17-0487
- Simonson DC, Vernon A, Foster K, Halperin F, Patti ME, Goldfine AB. Adjustable gastric band surgery or medical management in patients with type 2 diabetes and obesity: three-year results of a randomized trial. *Surg Obes Relat Dis*. 2019;15(12): 2052-2059. doi:10.1016/j.soard.2019.03.038
- Cummings DE, Arterburn DE, Westbrook EO, et al. Gastric bypass surgery vs intensive lifestyle and medical intervention for type 2 diabetes: the CROSSROADS randomised controlled trial.

Diabetologia. 2016;59(5):945-953. doi:10.1007/s00125-016-3903-x

20. Arterburn D, Flum DR, Westbrook EO, et al; CROSSROADS Study Team. A population-based, shared decision-making approach to recruit for a randomized trial of bariatric surgery versus lifestyle for type 2 diabetes. *Surg Obes Relat Dis*. 2013;9(6):837-844. doi:10.1016/j.soard.2013.05.006

21. Buse JB, Caprio S, Cefalu WT, et al. How do we define cure of diabetes? *Diabetes Care*. 2009;32(11):2133-2135. doi:10.2337/dc09-9036

22. Riddle MC, Cefalu WT, Evans PH, et al. Consensus report: definition and interpretation of remission in type 2 diabetes. *Diabetes Care*. 2021;44(10):2438-2444. doi:10.2337/dci21-0034

23. van der Wal WM, Geskus RB. ipw: an R package for inverse probability weighting. *J Stat Softw*. 2011;43(13):1-23.

24. Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. *Stat Med*. 2015;34(28):3661-3679. doi:10.1002/sim.6607

25. Hernán MA, Brumback BA, Robins JM. Estimating the causal effect of zidovudine on CD4 count with a marginal structural model for repeated

measures. *Stat Med*. 2002;21(12):1689-1709. doi:10.1002/sim.1144

26. Lean MEJ, Leslie WS, Barnes AC, et al. Durability of a primary care-led weight-management intervention for remission of type 2 diabetes: 2-year results of the DiRECT open-label, cluster-randomised trial. *Lancet Diabetes Endocrinol*. 2019;7(5):344-355. doi:10.1016/S2213-8587(19)30068-3

27. Knowler WC, Barrett-Connor E, Fowler SE, et al; Diabetes Prevention Program Research Group. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *N Engl J Med*. 2002;346(6):393-403. doi:10.1056/NEJMoa012512

28. Wing RR, Bolin P, Brancati FL, et al; Look AHEAD Research Group. Cardiovascular effects of intensive lifestyle intervention in type 2 diabetes. *N Engl J Med*. 2013;369(2):145-154. doi:10.1056/NEJMoa1212914

29. Lean ME, Leslie WS, Barnes AC, et al. Primary care-led weight management for remission of type 2 diabetes (DiRECT): an open-label, cluster-randomised trial. *Lancet*. 2018;391(10120):541-551. doi:10.1016/S0140-6736(17)33102-1

30. Coleman KJ, Haneuse S, Johnson E, et al. Long-term microvascular disease outcomes in

patients with type 2 diabetes after bariatric surgery: evidence for the legacy effect of surgery. *Diabetes Care*. 2016;39(8):1400-1407. doi:10.2337/dc16-0194

31. Rubino F, Nathan DM, Eckel RH, et al; Delegates of the 2nd Diabetes Surgery Summit. Metabolic surgery in the treatment algorithm for type 2 diabetes: a joint statement by International Diabetes Organizations. *Diabetes Care*. 2016;39(6):861-877. doi:10.2337/dc16-0236

32. Schiavon CA, Bersch-Ferreira AC, Santucci EV, et al. Effects of bariatric surgery in obese patients with hypertension: the GATEWAY randomized trial (Gastric Bypass to Treat Obese Patients With Steady Hypertension). *Circulation*. 2018;137(11):1132-1142. doi:10.1161/CIRCULATIONAHA.117.032130

33. Consensus Development Conference Panel. NIH conference: gastrointestinal surgery for severe obesity. *Ann Intern Med*. 1991;115(12):956-961. doi:10.7326/0003-4819-115-12-956

34. Eisenberg D, Shikora SA, Aarts E, et al. 2022 American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): indications for metabolic and bariatric surgery. *Surg Obes Relat Dis*. 2022;18(12):1345-1356. doi:10.1016/j.soard.2022.08.013